EV Technology Course — Course Map

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This is the second course in your EV programme. The battery course covered what stores the energy; this one covers everything else in an electric vehicle: how the vehicle is laid out, how each major component works, how they are controlled, tested, manufactured and costed, and which new technologies are changing them. It has 17 modules, and each one follows the same 16-part lesson format as the battery course.

#	Module	What you will be able to do
1	EV vehicle architecture and system overview	Trace power and signals through a car, a scooter and a bus; explain the skateboard platform
2	Electric traction motors	Explain PMSM, induction, BLDC, SRM, wound-rotor and axial-flux motors; read a torque–speed curve
3	Power electronics: inverter, DC-DC and onboard charger	Explain how 6 switches make AC from DC; compare IGBT, SiC and GaN
4	E-drive integration: gears, e-axles and NVH	Size a reduction ratio; explain 3-in-1 and multi-in-one drives and gear whine
5	Vehicle control: VCU, motor control and regenerative braking	Explain torque arbitration, field-oriented control, regen blending and traction control
6	HV and LV electrical system	Lay out HV distribution, contactors, harnesses, connectors, isolation and the 12/48 V network
7	Vehicle thermal management and HVAC	Explain heat pumps, chillers, integrated thermal modules and how heating affects range
8	Chassis, body, crash safety and lightweighting	Explain skateboard design, battery crash protection, castings and materials
9	Brakes, steering and suspension: by-wire systems	Explain brake-by-wire, electric power steering and steer-by-wire with redundancy

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10	E/E architecture, software and connectivity	Compare domain and zonal architectures; explain the software-defined vehicle, OTA and cybersecurity
11	Energy efficiency, range and consumption	Build a range model; quantify aero, tyre, HVAC and drivetrain losses
12	Testing, validation and homologation	Explain the V-model, dyno and HIL testing, EMC, crash and Indian certification (CMVR, AIS)
13	EV manufacturing and cost engineering	Compare EV and ICE plants and BOMs; build an e-drive should-cost
14	Electric two- and three-wheelers	Explain hub vs mid-drive, controllers, swapping and India-specific design choices
15	Commercial EVs, buses, trucks and fuel cells	Explain e-bus and e-truck design, charging strategies and the FCEV powertrain
16	New and emerging EV technologies	Rate 800–1,000 V, axial-flux, rare-earth-free motors, by-wire, wireless charging and more by maturity
17	Capstone: EV teardown and should-cost project	Run an end-to-end teardown, BOM and should-cost comparison of a compact electric SUV against its competitors

How the courses connect. Battery topics (cells, BMS, charging, recycling) are covered in the battery course and only referenced here. Modules 13 and 17 lean on your costing background (should-cost, BOM, VA/VE).